waves. While a particle can only pass through one slit, its pilot wave passes through both and interferes with itself. That self-interference, in turn, affects the motion of the particle, because it is guided by the wave. The wave pushes the particle onto a path ensuring the appearance of an interference pattern on the photographic plate after enough photons have been sent through the double slit (Figure 5.4). Putting photon detectors on each slit affects each photon's pilot wave—no matter how ingenious the design, any photon detector must alter a photon's pilot wave, as ensured by Heisenberg's uncertainty principle, which in Bohm's interpretation places limits on how much measuring devices can avoid interfering with the things they attempt to measure. The effect of these measurements on the photons' pilot waves alters their trajectories, causing them to form a pair of clusters on the photographic plate rather than an interference pattern. In Bohm's account, although measurement can influence a particle's motion, all particles have definite positions whether or not anyone is looking at them.

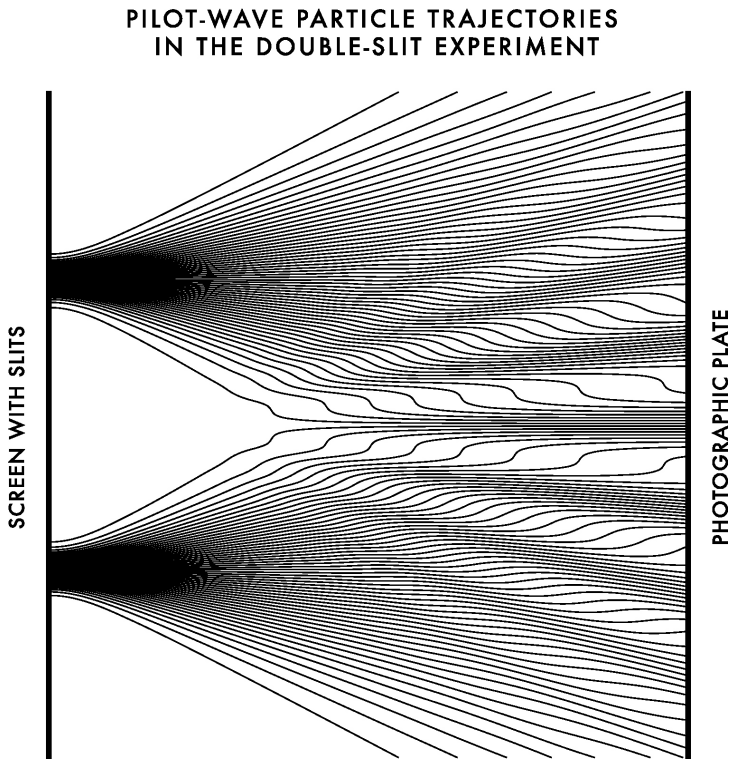


Figure 5.4. Particle trajectories guided by pilot waves in the double-slit experiment (top-down view). Figure produced with Mathematica code graciously provided by Professor Charles Sebens of UCSD.